

Dark Vessel Detection via Multimodal SAR & CGCAM

Computer Vision Project Presentation

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Presentation Outline

01

Problem Statement

02

**SAR Fundamentals &
Challenges**

03

State of the Art

04

**Proposed Method &
Architecture**

05

**CGCAM Novelty &
Deep-Dive**

06

**Dataset & Experimental
Setup**

07

**Model Evaluation &
Ablation Study**

08

**Conclusions & Future
Work**



Problem Statement

GLOBAL ECONOMIC LOSS

**\$10B –
\$23.5B**

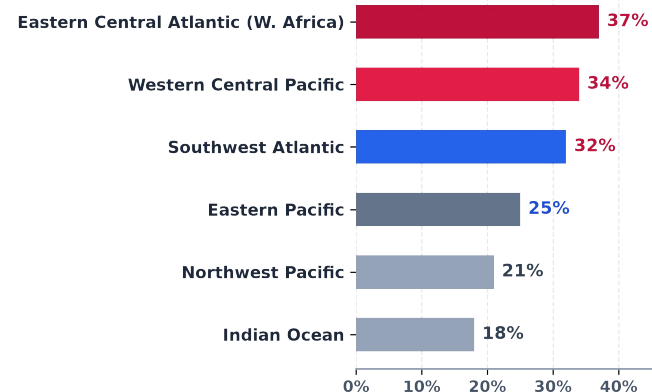
per year globally (FAO /
Agnew et al.)

UNMONITORED HARVEST

**1 in every
5**

fish caught illegally
(~20% global total)

Vulnerable Maritime Regions (IUU Catch Share)



COOPERATIVE VESSEL (AIS ON)

Flow: Merchant / Legal Vessel → Transmits AIS Radio Coordinates → Continuous 24/7 Global Maritime Tracking

Legal Vessel

AIS Broadcast

24/7 Monitored

DARK VESSEL (NON-COOPERATIVE)

Flow: Clandestine IUU Vessel → Disables Transponder (AIS Blackout) → Optical/Radio Blind spot → **SOLUTION: Sentinel-1 SAR Satellite Radar**

IUU Clandestine

AIS Blackout

Sentinel-1 SAR Radar

DARK VESSELS KEY DEFINITION

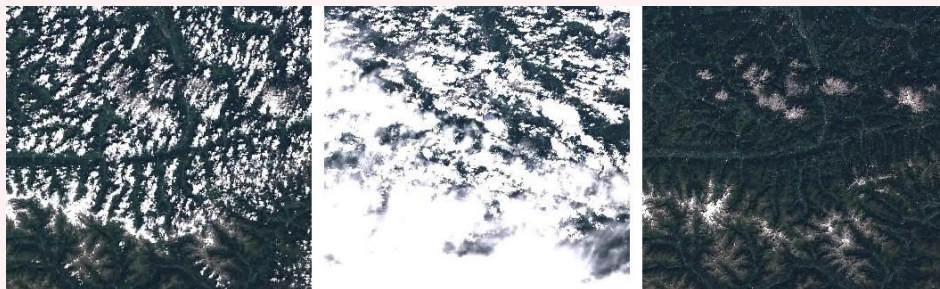
Vessels engaged in illegal fishing or illicit trafficking that intentionally disable their AIS. Completely invisible to optical cameras (blinded by clouds or night time), they are uniquely detectable through active microwave **Sentinel-1 SAR satellite radar**.



Problem Statement (2)

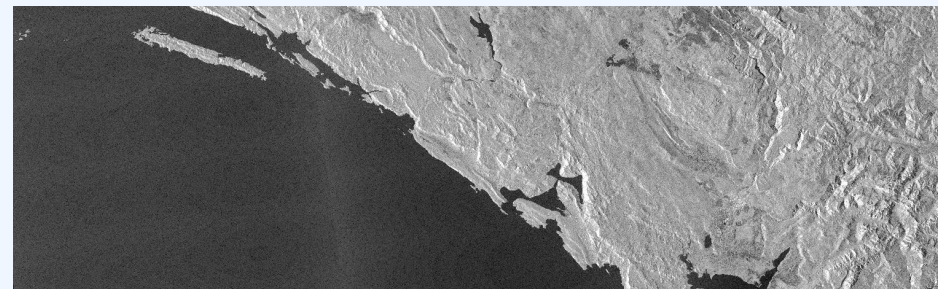
Why optical cameras fail in ocean monitoring and SAR radar is the only 24/7 operational modality

✗ PASSIVE OPTICAL SENSOR (RGB / VISIBLE)



- **Blind under Cloud Cover:** Over 72% of the global ocean surface is constantly obscured by clouds, fog, or rain (source: NASA/MODIS).
- **Zero Night Visibility:** Passive sensors depend entirely on solar illumination; nighttime hours (50% of operational time) are totally blacked out.
- **Ineffective for Dark Vessels:** Illicit maritime traffic systematically operates under dark hours and severe weather cover to evade detection.

✓ ACTIVE SAR RADAR (SENTINEL-1 C-BAND)



- **Active C-Band Microwaves (5.405 GHz, $\lambda \approx 5.54$ cm):** Emits synthetic radar pulses that penetrate weather, clouds, and storms completely unhindered.
- **All-Weather, Day & Night Operation:** Guarantees 24/7 continuous surveillance capability, fully independent of solar light and meteorological state.
- **High Metallic Contrast:** Metallic hulls generate intense dihedral double-bounce scattering, standing out sharply against sea clutter.

TAKEAWAY: SAR imaging is the only globally reliable satellite approach for detecting dark vessels. However, the total absence of visual texture, severe speckle noise, and coastal false alarms necessitate specialized Computer Vision architectures (CGCAM CenterNet).

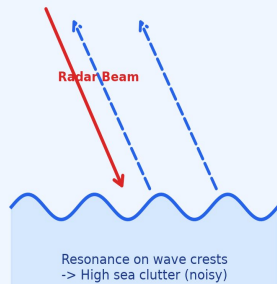
SAR Fundamentals & Sensor Challenges

Physics of Sentinel-1 C-Band Radar and the three core failure modes of conventional detectors

1. POLARIMETRY (VV vs VH)

Bragg Scattering (Sea)

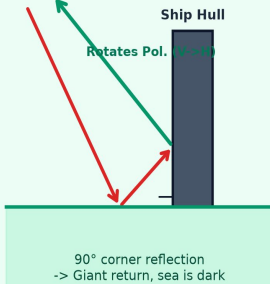
Co-Polarized (VV Channel)



TCR is LOW

Dihedral Double-Bounce

Cross-Polarized (VH Channel)



TCR(VH) >> TCR(VV)

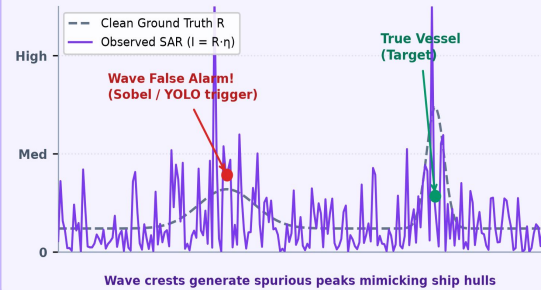
- **VV Channel (Co-polarized):** Bragg resonance with wind ripples bright sea surface, high sea clutter.
- **VH Channel (Cross-polarized):** Hull-sea double bounce sea near noise floor, ship extremely bright.
- **Signal-to-Clutter Ratio:** TCR(VH) >> TCR(VV), critical for discriminating metal from ocean clutter.

2. MULTIPLICATIVE SPECKLE NOISE

$$I(x, y) = R(x, y) \cdot \eta(x, y)$$

Multiplicative Noise: $\eta \sim \Gamma(L, 1/L)$ ($E = 1$, $Var = 1/L$)

Noise scales with signal brightness: bright waves fluctuate wildly!

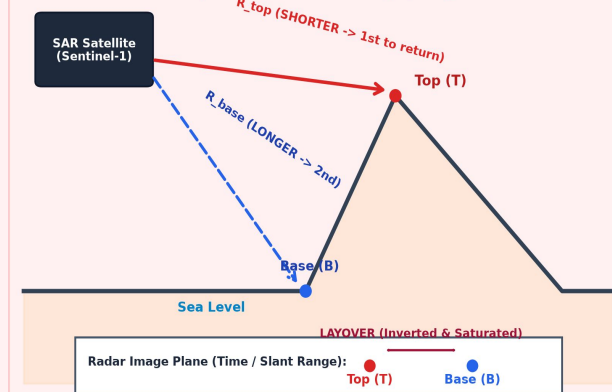


- **Multiplicative Nature:** $I = R$ where $\Gamma(L, 1/L)$; noise scales proportionally with signal intensity.
- **Severe Fluctuations:** High-return zones oscillate heavily; traditional gradient filters (Sobel, Canny) fail.
- **Detector Failure:** Wind-driven breaking wave crests are misclassified as ship hulls high false alarm rates.

3. COASTAL GEOMETRY (LAYOVER)

Radar Slant-Range & Layover

Cliff Top returns before Base: $R_{top} < R_{base}$



- **Slant-Range Geometry:** Radar measures oblique time-of-flight; cliff tops appear closer to sensor than base.
- **Geometric Compression:** Coastal terrain collapses toward sensor (layover), creating blinding white masks.
- **Shoreline Blindness:** Bright coastal rocks overpower metallic hulls causes thousands of false alarms.

METHODOLOGICAL BRIDGE: Standard detectors fail on sea clutter and coastal cliffs. Eliminating these false alarms requires a geophysical guide: our **CGCAM module**.



State of the Art & Conventional Detector Limitations

Why statistical algorithms, standard optical detectors, and early concatenation fail in maritime SAR

1. STATISTICAL METHODS (CFAR)

Algorithms: CA-CFAR, OS-CFAR, 2-Parameter CFAR (Local clutter estimation)

- **Operational Principle:** Estimate ocean noise distributions (K/Weibull) over sliding windows to guarantee a fixed false alarm rate (Pfa).
- **Homogeneity Assumption:** Require calm, uniform seas; completely collapse under rough seas or high winds (non-stationary Bragg clutter).
- **Target Masking & Coasts:** Closely spaced targets shadow each other (target masking), while rocky cliffs destroy statistical estimates.

2. STANDARD DEEP LEARNING (YOLO)

Models: Faster R-CNN, RetinaNet, YOLOv5/v8 (Designed for natural RGB)

- **Lack of Visual Texture:** Trained on RGB colors and edges; highly vulnerable to multiplicative SAR speckle noise ($I = R \cdot \eta$).
- **Wave & Coastal False Alarms:** Misclassify wave crests as ship hulls and get blinded by severe layover artifacts along rocky shorelines.
- **Anchor Box Inefficiency:** Predefined bounding boxes fail for tiny vessels (few pixels) arbitrarily oriented in vast open ocean scenes.

3. EARLY CONCATENATION

Naïve Approach: Concatenating SAR + Context directly in the 1st Conv2D layer

- **Spatial & Physical Mismatch:** Abruptly merges 10 m/px high-res radar with low-frequency geophysical fields (bathymetry/wind at ~500 m/px).
- **Radar Gradient Degradation:** Early convolutional filters get confused, degrading subtle double-bounce scattering patterns from ship hulls.
- **Scientific Evidence (Bai et al., 2024):** Literature demonstrates simple early fusion yields no performance gain over SAR-only baselines.

THE RESEARCH GAP: An anchor-free detector is required for sparse targets, combined with an architectural decoupling where context guides radar features: the CGCAM module.

Data Ingestion, Modalities & Pipeline Architecture

Multimodal origin, high-performance streaming via memory-mapping, and preprocessing contracts

1. DATA SOURCE & MODALITIES

Source: xView3-SAR Benchmark (Global Fishing Watch / DIU / ESA Copernicus)

- **Primary Radar Channels (10 m/px):** High-resolution Sentinel-1 C-band dual-polarization: VH (cross-pol) and VV (co-pol) at 16-bit float.
- **Auxiliary Geophysical Priors (~500 m/px):** Global ocean bathymetry model (GEBCO) and surface wind speed fields (OWI).
- **Semantic Ground Truth:** Maritime centroids annotated by cross-referencing AIS and satellite tracking for mobile vessels and fixed offshore structures.

2. EFFICIENCY & PREPROCESSING

Architecture: Zero-RAM Streaming via numpy.memmap on giant GeoTIFFs

- **Memory-Mapped Streaming:** 22k × 30k pixel scenes (>1.3 GB per channel). The pipeline maps disk files to memory, cropping patches on-demand without RAM saturation.
- **Multi-Resolution Alignment:** Bilinear resampling of low-frequency geophysical fields (500 m) to match the metric SAR grid (10 m).
- **Radiometric Normalization & Nodata:** Independent dB calibration (VH [-35, +10], VV [-30, +15]); sensor nodata (-32768) masked prior to normalization.

3. ROBUSTNESS & DATASET CONTRACT

Guaranteed Output: Batch (B, 5, 256, 256) in [0.0, 1.0] for DarkVesselNet

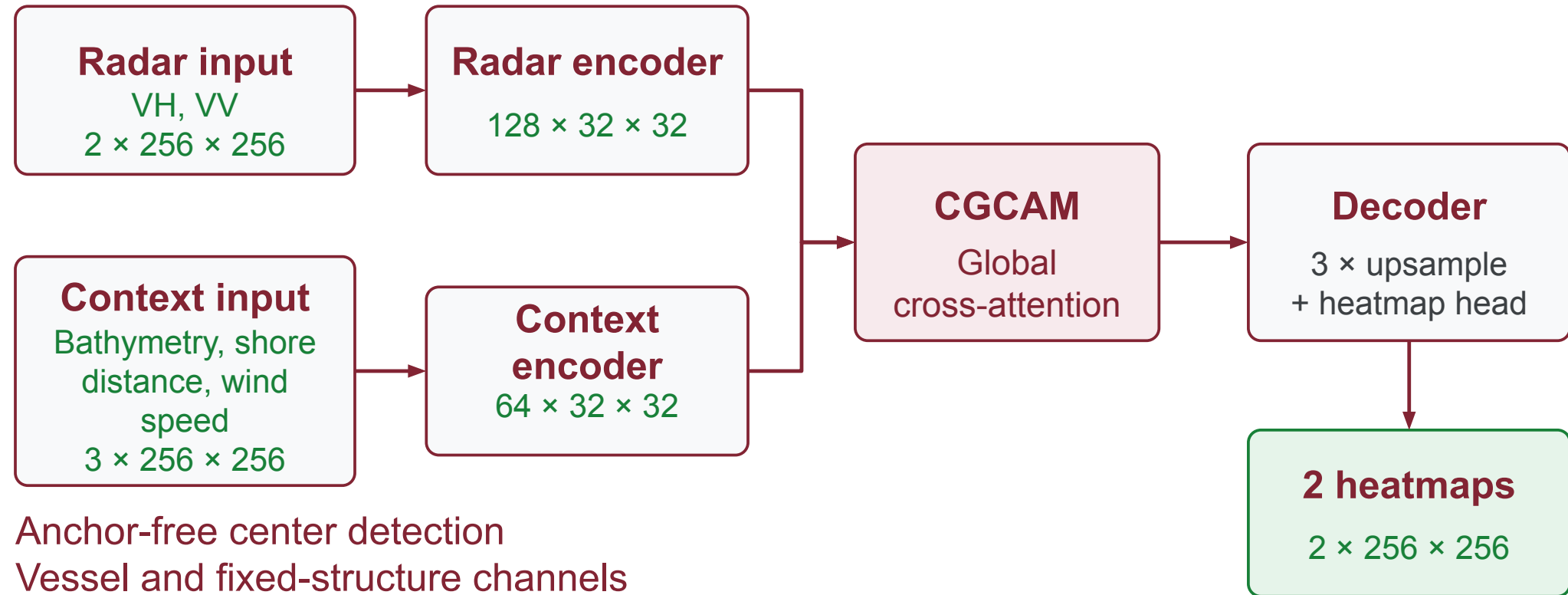
- **Derived Shore Channel:** Shore distance computed via Euclidean Distance Transform (EDT) from water mask (MAE = 1.15 km).
- **50:50 Balanced Sampling w/ Jitter:** Patches centered with ±108 px jitter (breaks center-shortcut bias) balanced with rocky coastline hard negatives.
- **Rigorously SAR-Safe Augmentation:** Exclusively 90° rotations and synchronous flips between inputs and Gaussian targets (scales/shears strictly forbidden).

FINAL CONTRACT TENSOR: Input (B, 5, 256, 256) -> Ground Truth (B, 2, 256, 256).



DarkVesselNet architecture

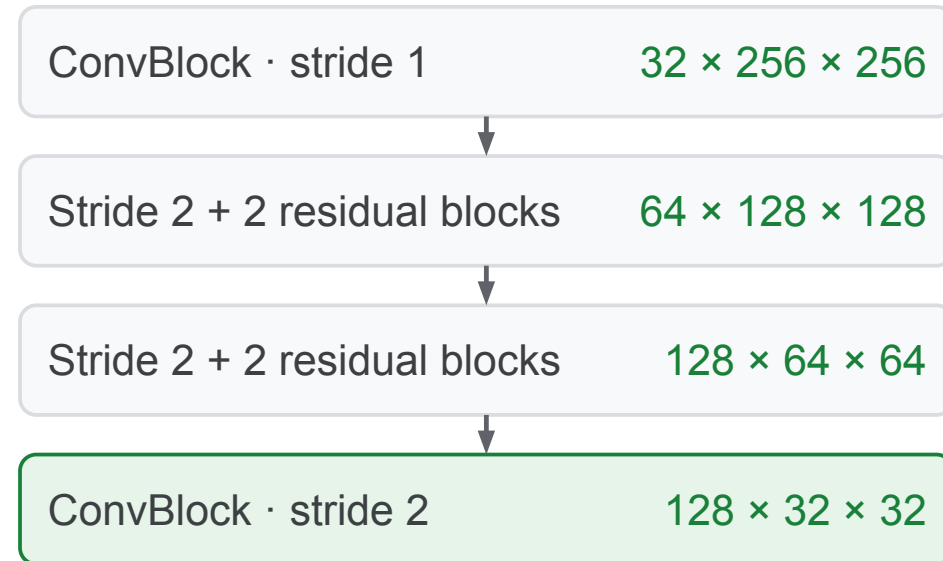
Separate radar and context streams, fused at 32×32 .



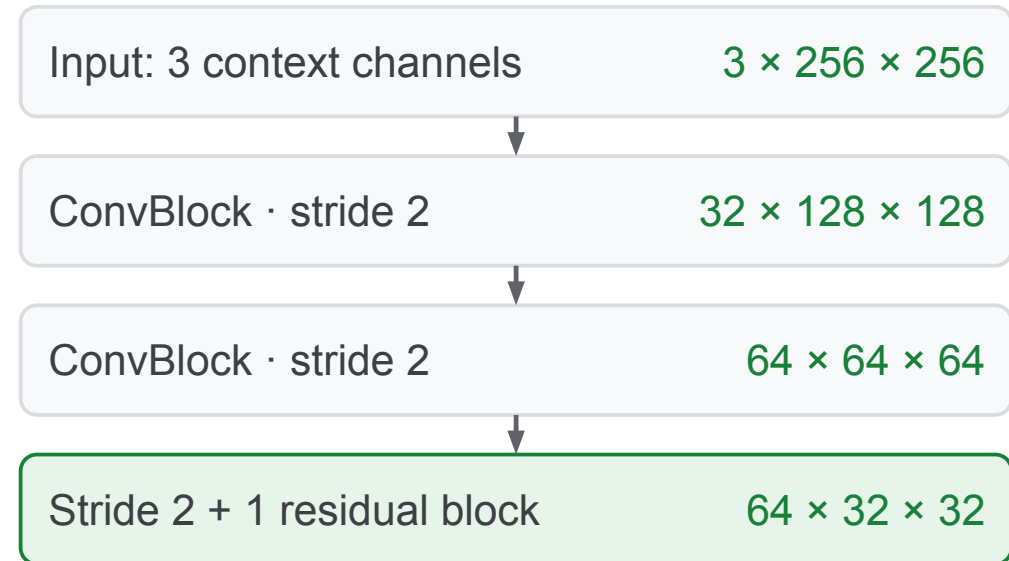
DarkVesselNet architecture(2)

Three downsampling stages produce matching spatial resolution.

Radar encoder



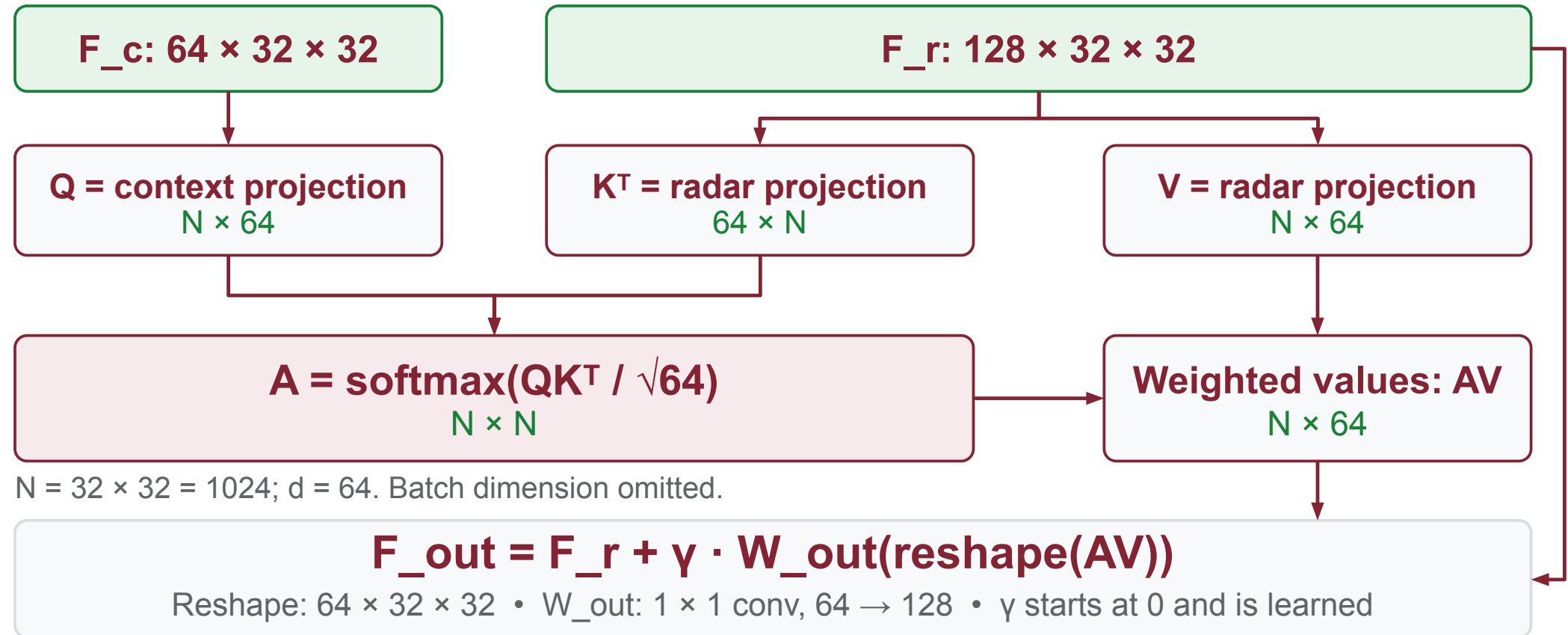
Context encoder



ConvBlock = 3×3 convolution + BatchNorm + ReLU
ResidualBlock = two 3×3 convolutions + identity skip

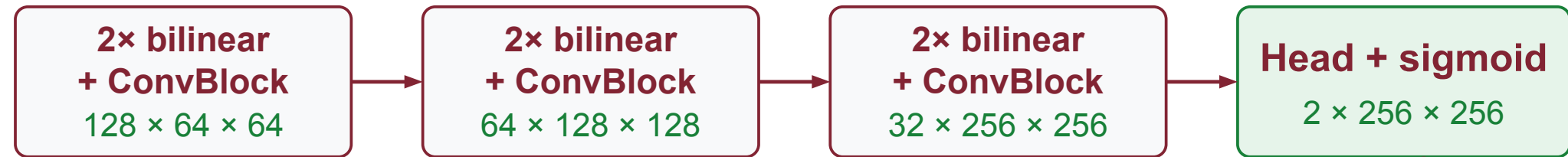
DarkVesselNet architecture(3)

Context supplies queries; radar supplies keys and values.



DarkVesselNet architecture(4)

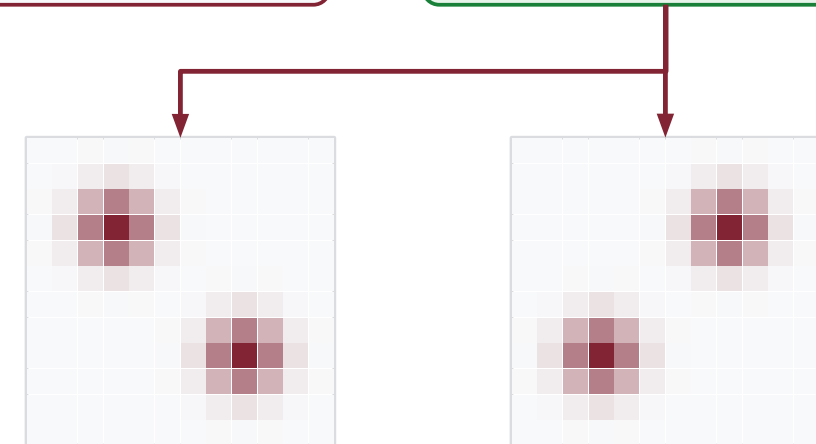
Bottleneck: $128 \times 32 \times 32 \rightarrow$ full-resolution center heatmaps.



From heatmaps to detections

1. Find 3×3 local maxima.
2. Apply the score threshold.

One threshold per model/run,
applied to both class channels.



Vessels

Structures

Schematic heatmaps · same spatial patch

Threshold selected on validation vessel F1, then fixed for test evaluation.

Dataset & Experimental Setup

7 scenes: 4 train / 1 val / 2 test.

Scene-level split, so no spatial leakage

Labels: xView3 CSV (confidence, **is_vessel**, distance-from-shore)

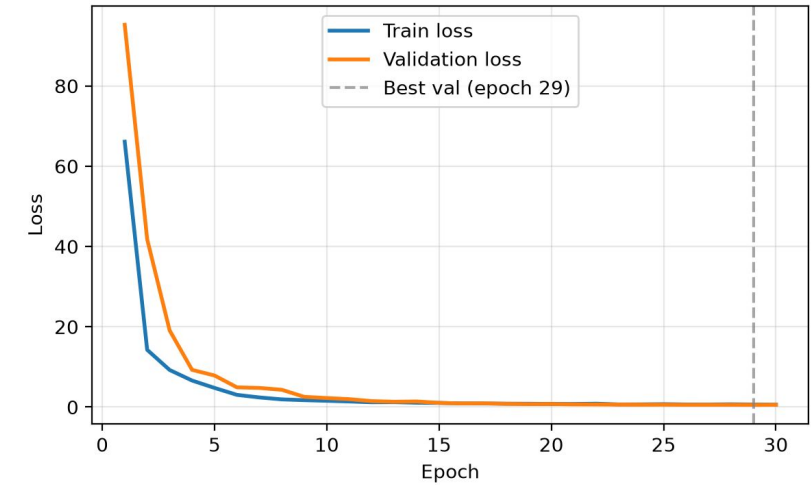
256×256 patches,
50:50
positive/negative

Loss: modified focal ($\alpha=2$, $\beta=4$) on Gaussian-rendered targets

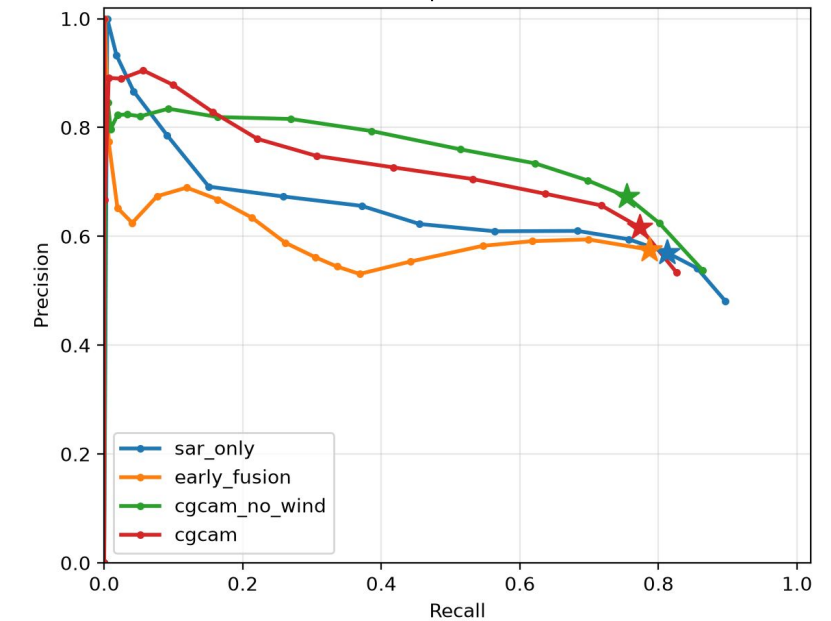
Validate
Vessel F1 on the validation scene after every epoch

Optuna: 5 optimizers × 4 variants → each variant uses its own tuned optimizer / LR / weight decay

Training convergence: cgcam



Precision-recall comparison: Vessel detection



Model evaluation & Ablation study

Variant	Precision	Recall	F1	False pos.	Near-shore FP	Near-shore share
SAR only	0.626 ± 0.013	0.603 ± 0.069	0.613 ± 0.037	$2,243 \pm 251$	760 ± 137	0.34 ± 0.02
Early fusion	0.579 ± 0.030	0.347 ± 0.091	0.430 ± 0.074	$1,561 \pm 293$	483 ± 86	0.31 ± 0.04
CGCAM, no wind	0.690 ± 0.040	0.465 ± 0.111	0.550 ± 0.092	$1,286 \pm 267$	467 ± 121	0.36 ± 0.08
CGCAM	0.689 ± 0.025	0.474 ± 0.075	0.559 ± 0.052	$1,343 \pm 281$	505 ± 127	0.37 ± 0.03

Fewer false alarms

CGCAM cuts total false positives by 40% ($2,243 \rightarrow 1,343$) and raises precision $0.63 \rightarrow 0.69$.

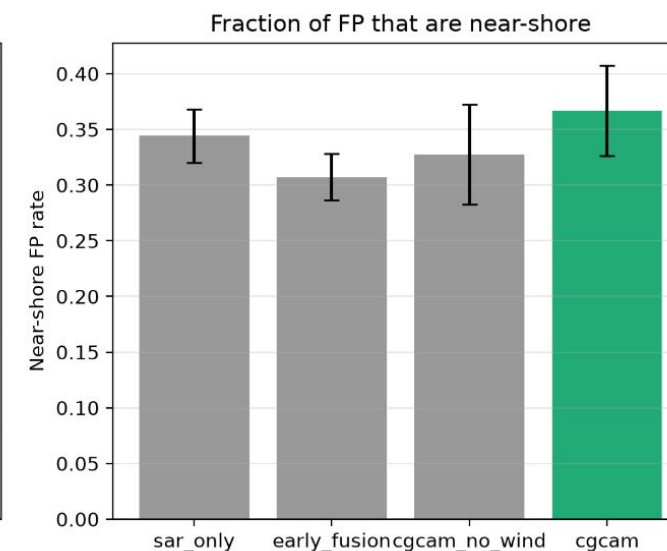
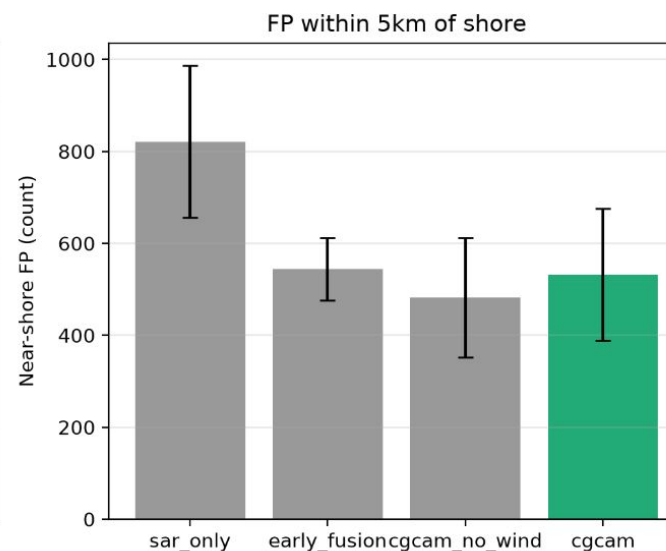
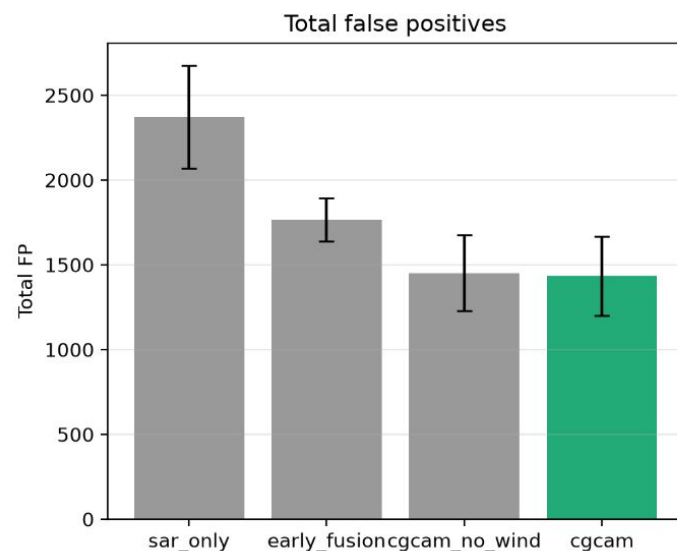
...but lower recall

Recall drops $0.60 \rightarrow 0.47$, so at the validation-chosen threshold F1 is lower ($0.61 \rightarrow 0.56$). SAR only has the best mean F1.

Not a coastal effect

Near-shore false positives fall in number ($760 \rightarrow 505$) but their share of all FP does not ($0.34 \rightarrow 0.37$).

Model evaluation & Ablation study (2)



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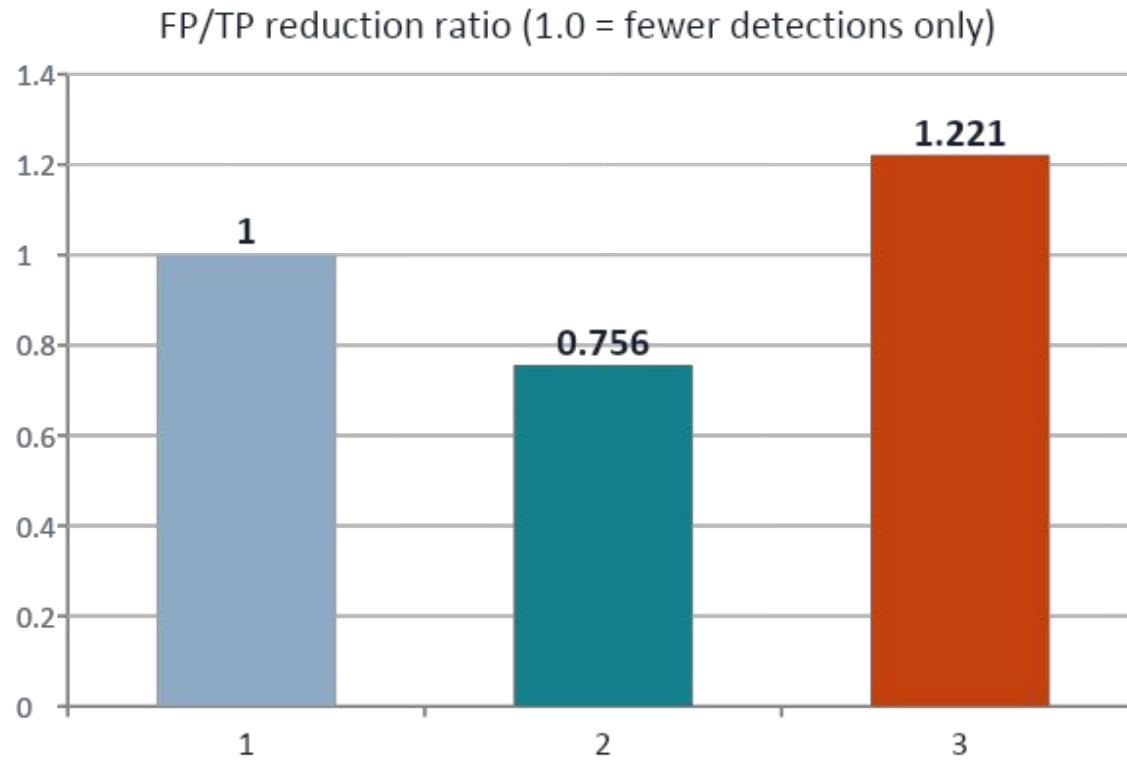
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Model evaluation & Ablation study (3)

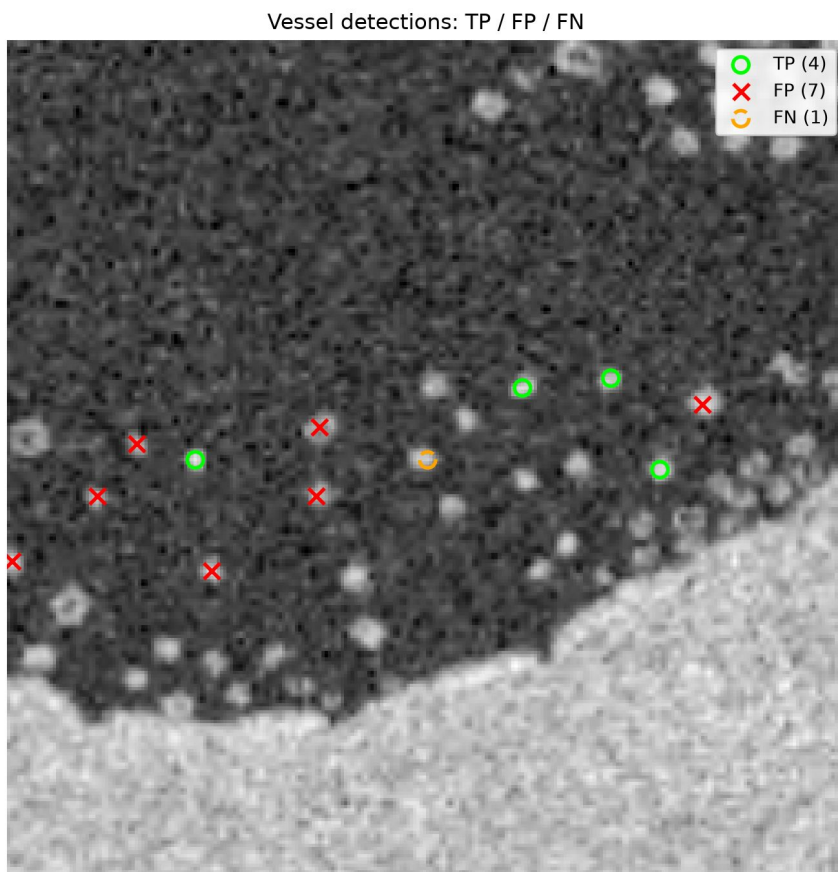


433

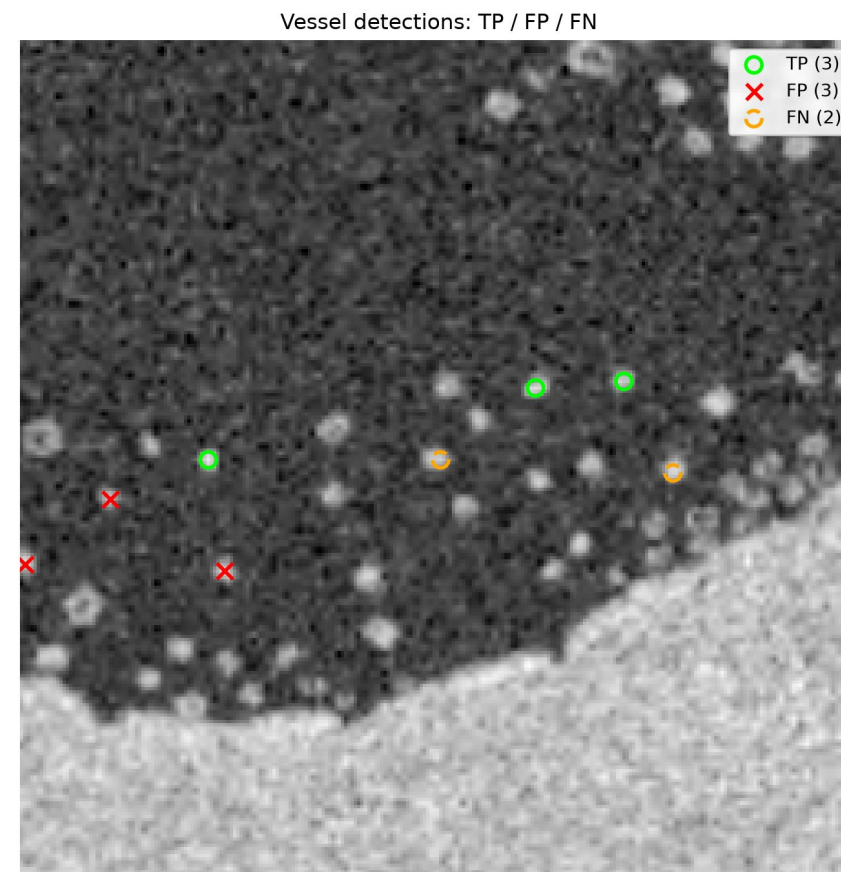
false alarms fewer than
detection volume alone
explains (-24%)

**CGCAM removes false positives
faster than true positives, so it is
more selective.**

Model evaluation & Ablation study (5)

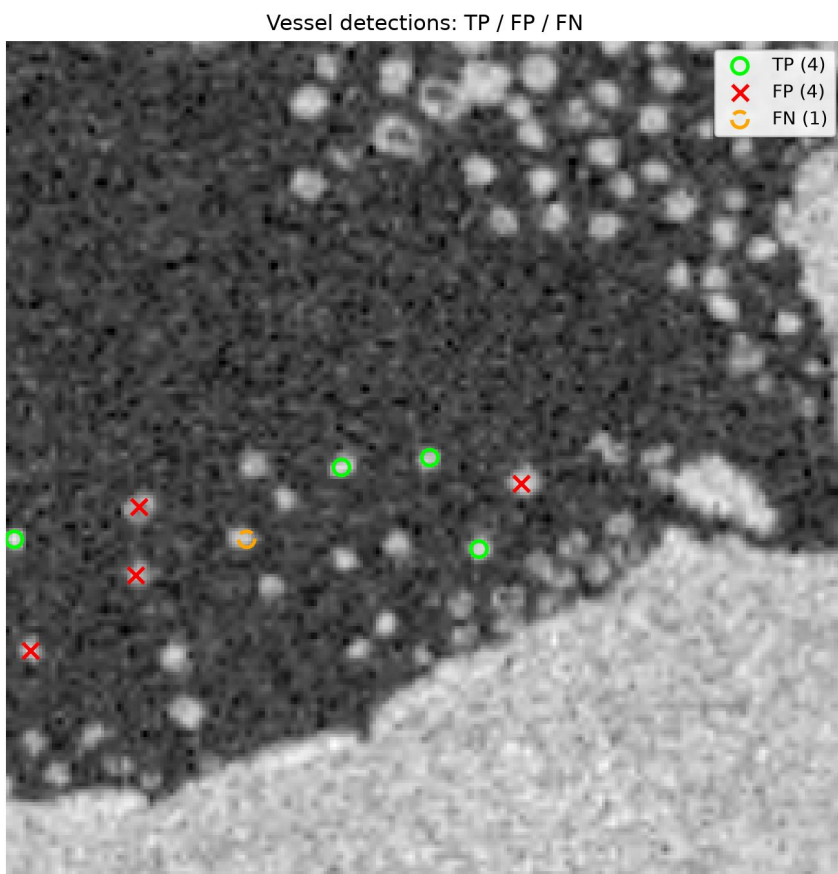


SAR_ONLY

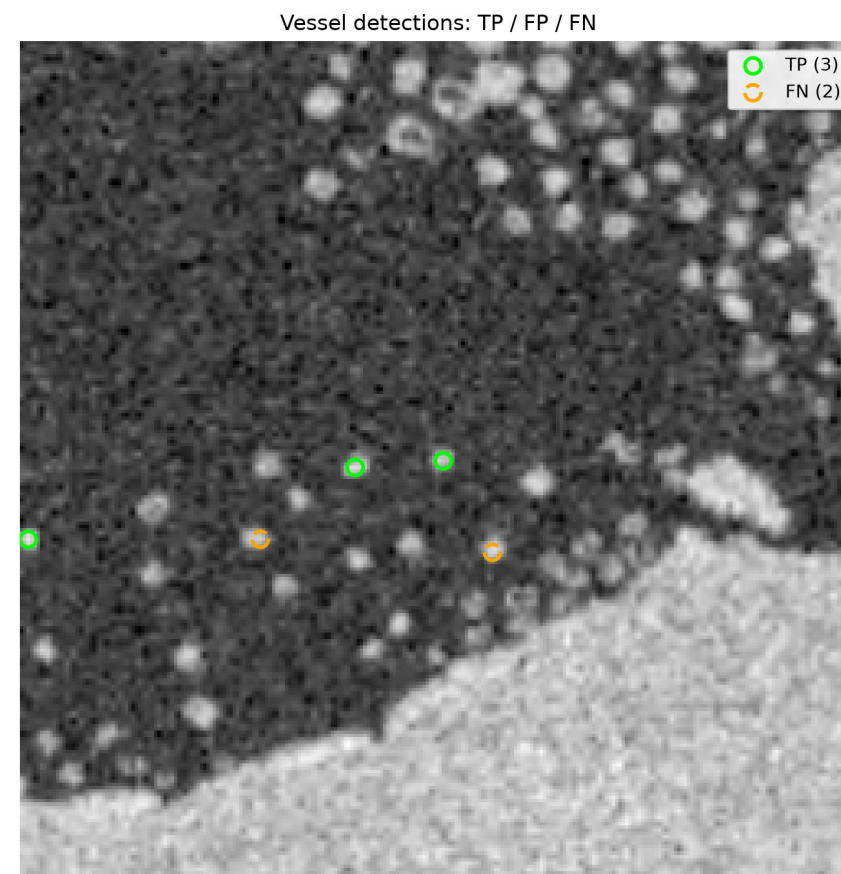


CGCAM

Model evaluation & Ablation study (4)



SAR_ONLY



CGCAM

Conclusions & Future Work

Conclusions

Context makes the detector more selective:

→ -40% false positives (2243→1343)
more trustworthy detections

→ Higher precision
63 → 0.69

The trade off is recall:

→ Recall 0.60 → 0.47 some vessels missed so F1 is lower [0.61→0.56]

Future work

Use of more data and validation scenes:

→ single validation scene is noisy:

- tuning and model selection rest on one scene
- overestimates performance (val F1 0.95, test 0.46-0.64)

Improve the precision recall trade-off:

→ scaling the gate with distance so it fades out offshore. Open water would behave like SAR-only, and near shore would get the context benefit.